

Phase Unwrapping and 1D Sign Problems

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Overview

- 1 Sign/Phase Problems in Finite Density
- 2 Phase Unwrapping
- 3 Toy Model Results: 0+1D Scalar Theory
- 4 Large Phase Jumps
- 5 Conclusions and Outlook

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Sign/Phase Problems in Finite Density

1. Complex action ($\mu \neq 0$):

$$Z = \int \mathcal{D}[\phi] e^{-S[\phi]}$$

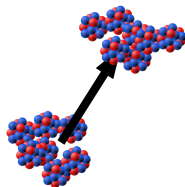
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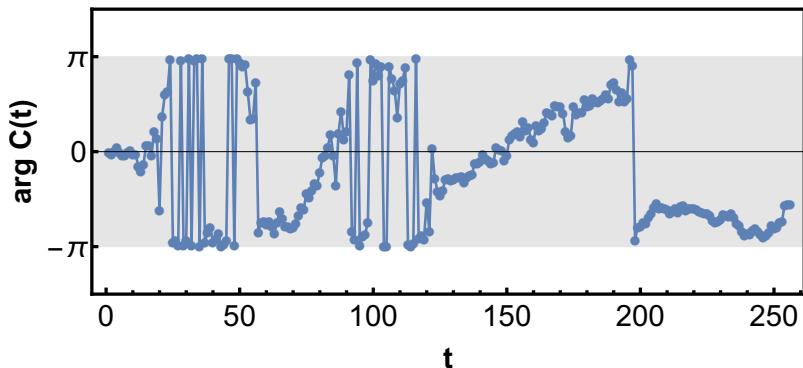
2. Complex observables ($C(t, 0)$ for $U(1)_B \neq 0$) **[our focus]**:

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int \mathcal{D}[\phi] \mathcal{O}[\phi] e^{-S[\phi]}$$



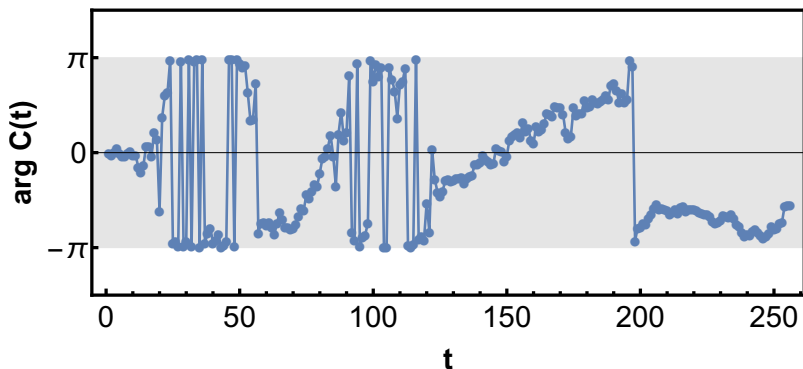
Sign problem = Signal-to-noise problem for correlators

Correlator phases drift relatively smoothly on typical configurations.



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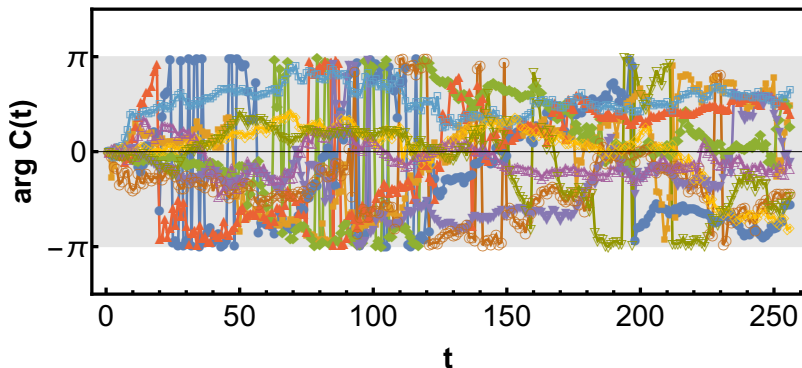
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Distributions of $\arg C(t)$ across the ensemble become increasingly uniform at large t .

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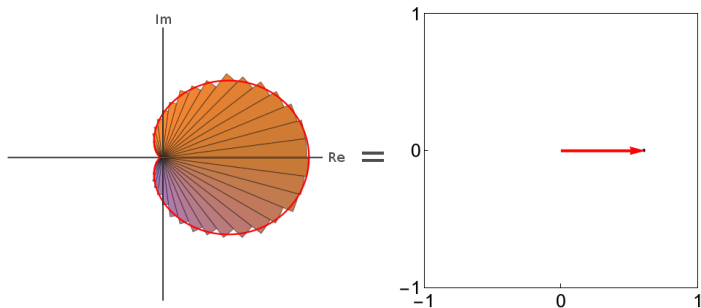
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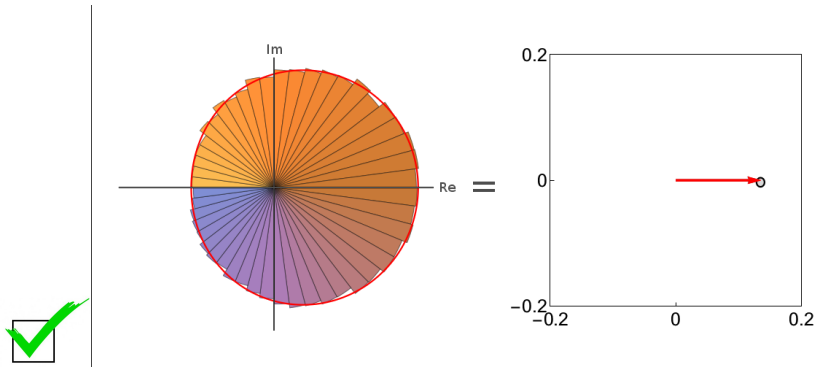
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Estimating $\langle C(t) \rangle$ at large t requires **exponentially precise cancellations** of nearly-uniform phase distributions.



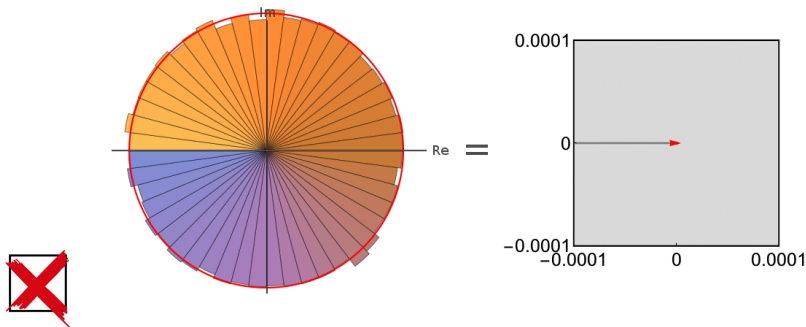
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Phase Unwrapping in 0+1D

1. Decompose $C_i(t)$ on each config i into phase $\Theta_i(t)$, log magnitude $\mathcal{R}_i(t)$.

2. “Unwrap” phase

$$\tilde{\Theta}_i(t) = \Theta_i(t) + 2\pi\nu_i(t).$$

3. Cumulant expansion allows precise estimate of $\langle C(t) \rangle$ by

$$\langle e^{\mathcal{R}(t)+i\Theta(t)} \rangle = \langle e^{\mathcal{R}(t)+i\tilde{\Theta}(t)} \rangle = e^{\sum_n \kappa_n/n!}.$$

[M G Endres, et al. PRL 107 (2011) 201601]

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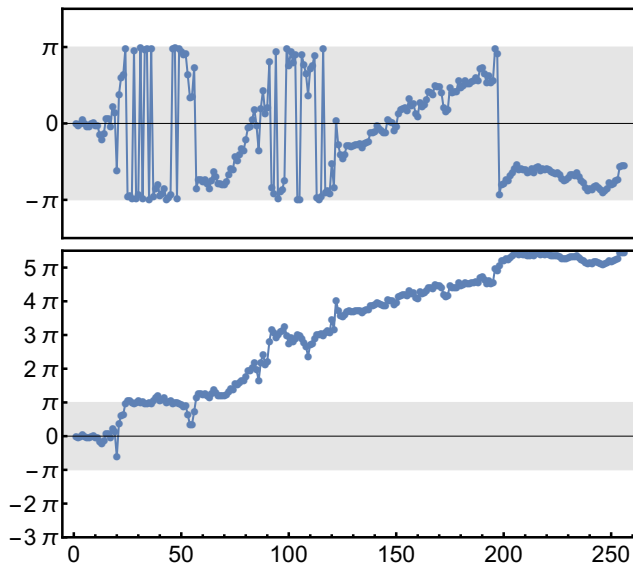
[M G Endres, et al. PRL 107 (2011) 201601]

For nearly-Gaussian, κ_2 dominates. Truncate $\sum_n$ at sufficiently large $n_{\max}$ to get a good estimate.

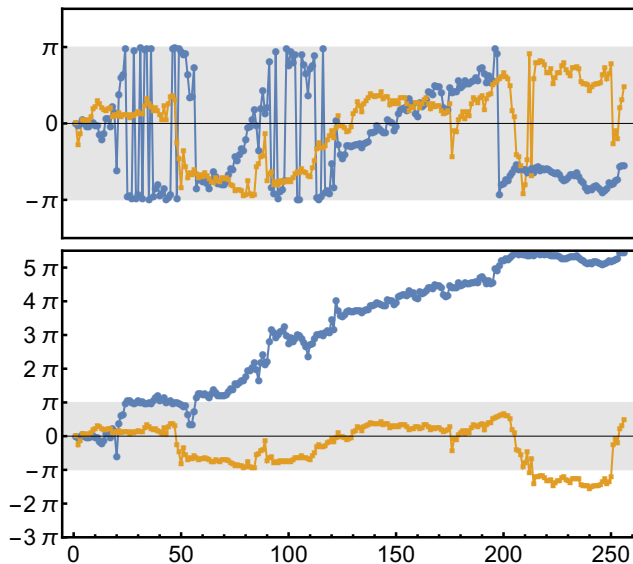
Unwrapping Phases

- ▶ Add $2\pi\nu$ to each $\Theta_i(t)$ according to a smoothing criterion (“unwrapping scheme”).
- ▶ Local iterative unwrapping: $\Theta_i(0) = 0$, unwrap forwards/backwards around half the lattice each way.
 - ▶ **Nearest-neighbor scheme:** minimize difference between previous unwrapped and current point.
 - ▶ **Window-based scheme:** minimize magnitude of differences between average of w previous unwrapping phases and current point.

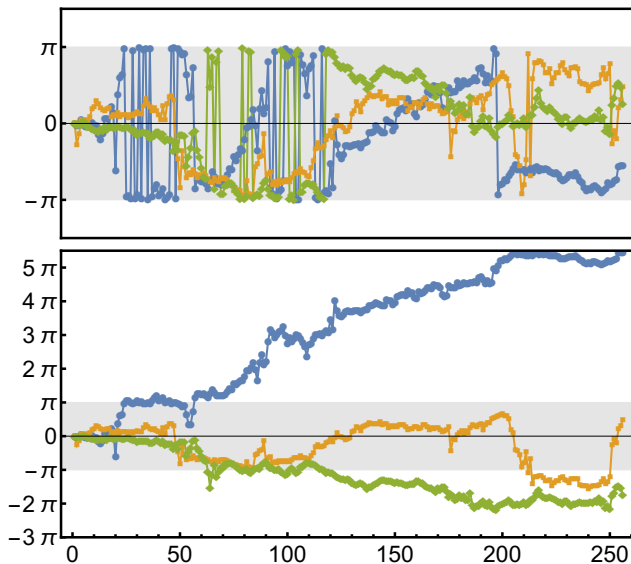
Nearest-Neighbor Example



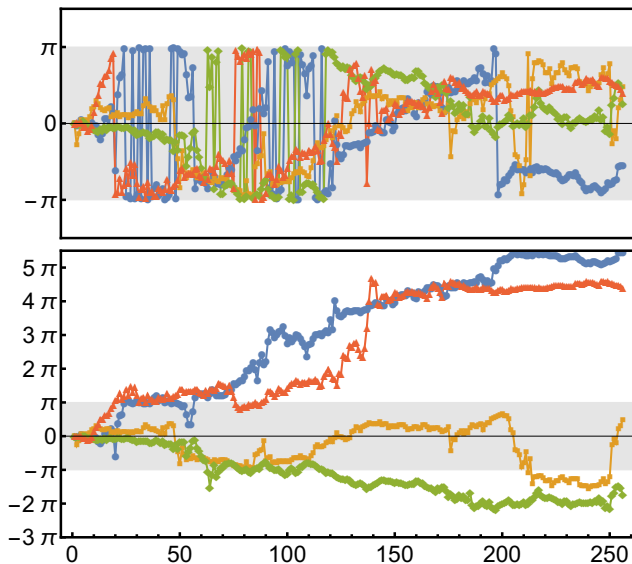
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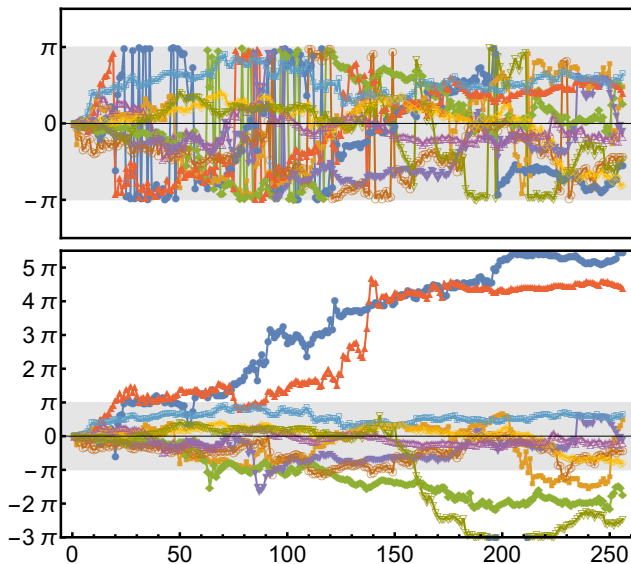
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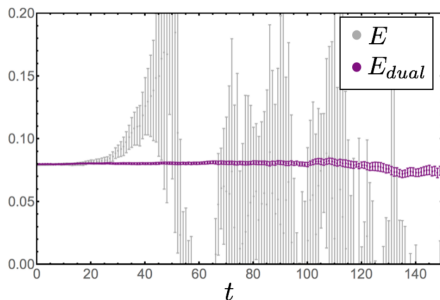


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0+1D Scalar Theory

$$S = \sum_{t=0}^{L-1} |\phi(t) - \phi(t-1)|^2 + M^2 |\phi(t)|^2 + \lambda |\phi(t)|^4$$

- ▶ Global $U(1)$ symmetry $\Rightarrow$ spectrum splits into charge sectors.
- ▶ Dual variables solution removes sign problem, giving a comparison. [M G Endres, PRD 75 (2007) 065012]



$$Z = 2 \sum_{q \in \mathbb{Z}} \int_0^\infty \prod_{t=0}^{L-1} \left[d|\varphi(t)| |\varphi(t)| e^{-2|\varphi(t)|^2 - V(|\varphi(t)|)} 2 I_{|q|}(\kappa(t)) \right]$$

0+1D Scalar Theory

Correlators experience maximally poor signal-to-noise:

$$\text{Sig} = \langle \phi^Q(t) \phi^{*Q}(0) \rangle \sim e^{-EQt}$$

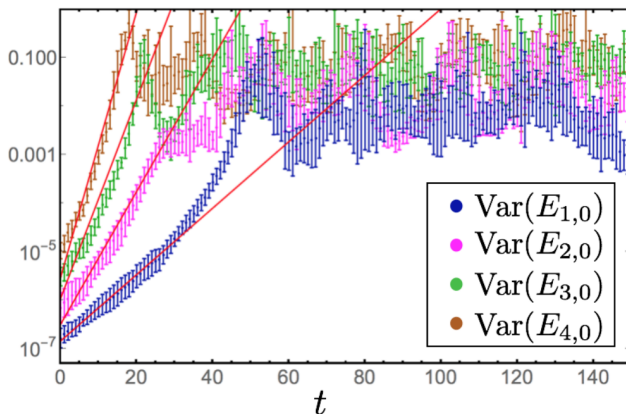
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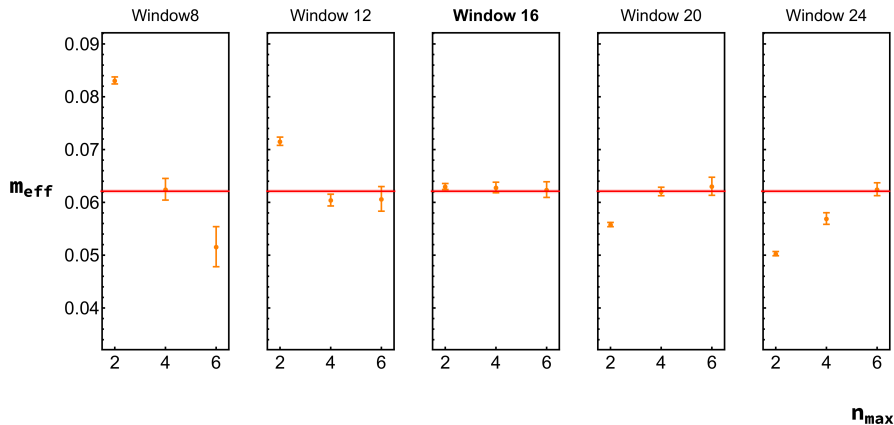
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$Q = 1$ Phase Unwrapping

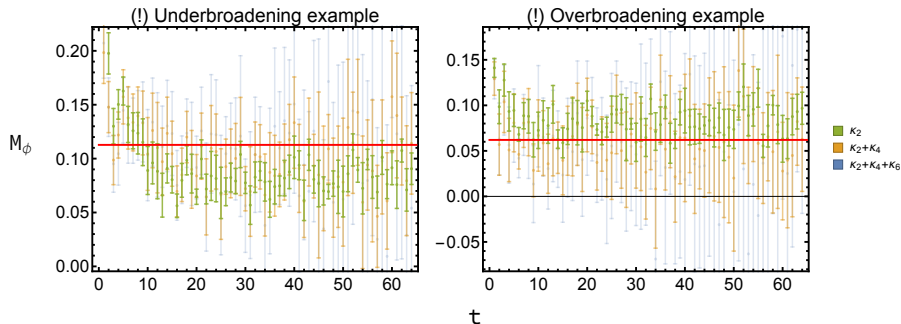
Per ensemble, unwrapping window w tuned to minimize higher-order cumulant corrections.

$$w \sim \xi \approx 16.1$$

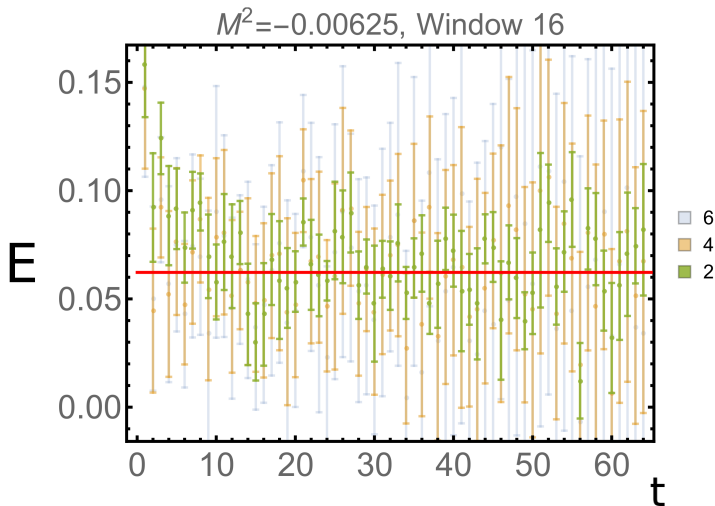


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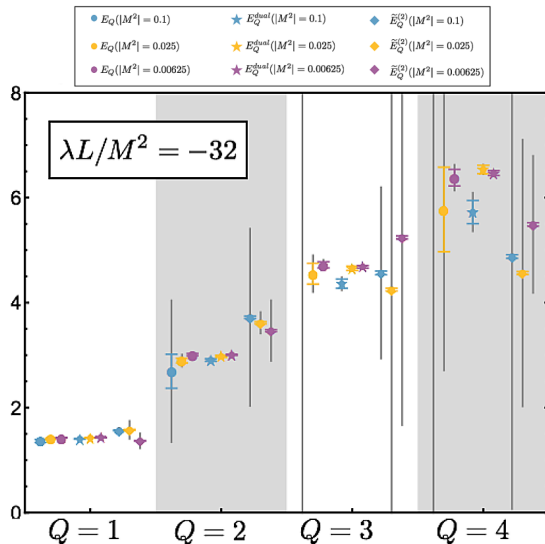
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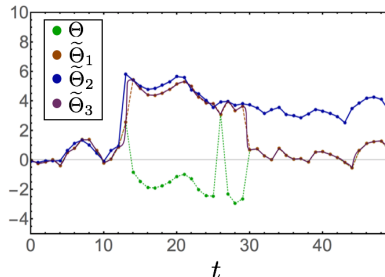
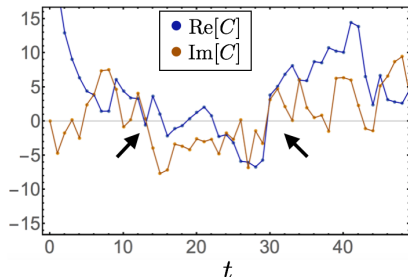
Spectrum Results



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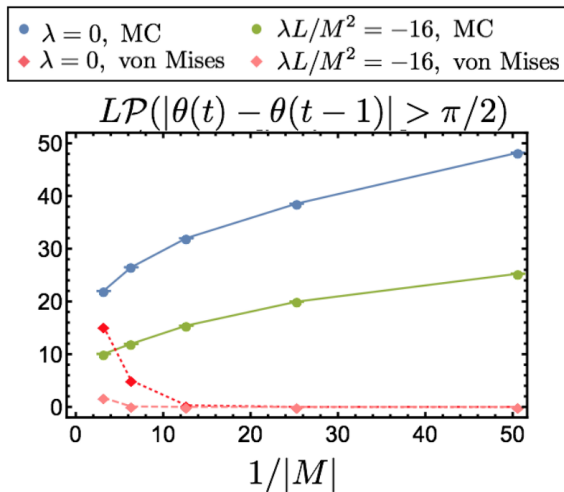
Large phase jumps cause scheme sensitivity

- ▶ Slowly drifting phases: insensitive to scheme.
- ▶ Quickly drifting phases: 2π differences in all subsequent points.



Large phase jumps cause scheme sensitivity

Frequency of phase jumps increase!



Sensitivity in higher dimensions

- ▶ 2D and higher enables unwrapping along multiple paths.
- ▶ Scheme dependent fluctuations in regions of unwrapped phase require large phase jumps along entire surface.

Unlike in 0+1D, expect no $O(V)$ fluctuations in the $a \rightarrow 0$ limit for higher dimensions. Scheme dependence might be removable as a discretization effect.

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2. Unwrapping performs an unbiased transform of the distribution:

$$\langle C(t) \rangle \equiv \langle e^{\mathcal{R}(t)+i\Theta(t)} \rangle = \langle e^{\mathcal{R}(t)+i\tilde{\Theta}(t)} \rangle$$

3. Cumulant expansion results in large scheme-dependent systematic uncertainties in 0+1D.
 - ▶ Replace cumulant expansion with reweighting approach
 - ▶ Work in higher dimensions

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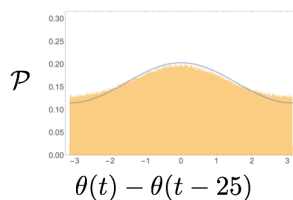
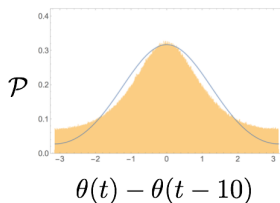
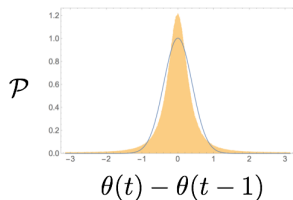
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Questions?

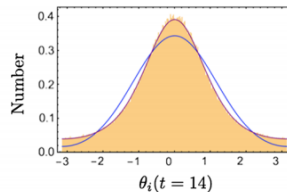
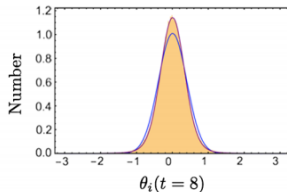
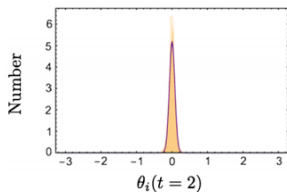
Backup Slides

How Gaussian?

0+1D Scalar Theory



LQCD Baryons [Wagman and Savage, PRD 96 (2017) 114508]



Multi-dim Outlook

Topological defects need to be considered. Real physics or simply a technical pathology of compact variables?

